Session 6 (Live demos): sample size, comparing groups, models and missing data

Foundations of Clinical Research · Week 6, Saturday 19:30 to 22:30 (online) · Neudata · instructor script

	Part A: sample size	Part B: compare groups	Part C: models and missing data
Slot	6.1, after the slide *Sample size: bigger effect, less noise* (about 20:05)	6.2 (about 20:21)	6.4, while showing its slides (21:30 to 21:48)
Duration	5 min (Steps 2, 3 and one change; full version 8 min if time allows)	6 min	woven into the 6.4 slides (about 8 min of R in total, optional)
Tool	free online calculator (e.g. OpenEpi) + `Course/Resources/tools/sample_size_calculator.xlsx`	`Course/Data/R/03_compare.R`	`Course/Data/R/04_models.R`, `05_missing.R`

Participants **watch your shared screen and predict in the chat**. They never calculate or run code. FLUID-ICU is **simulated**, and all its numbers below come from `Course/Data/key_results.md`.

Before the session

- [ ] Browser tab on the online calculator, zoomed to 150%, with the spreadsheet open on sheet `3_Means` as fallback.
- [ ] `Course/Data/FLUID-ICU.Rproj` open. Run `03_compare.R`, `04_models.R` and `05_missing.R` once end to end the night before (`05_missing.R` runs 20 imputations: allow about a minute).
- [ ] Fallback figures open: `Course/Resources/figures/en/data/` (`crude_vs_adjusted.png`, `forest_adjusted.png`, `km_balance.png`, `missing_lactate.png`, `cc_vs_mi.png`) (the comparison table is now a native slide table, *FLUID-ICU: four outcomes, four comparisons*).
- [ ] A slide or shared document listing the four sample-size inputs (expected effect, noise (SD), alpha, power) plus drop-out.

Part A: Sample size in 5 minutes (online calculator + spreadsheet)

In the 5-minute version run Steps 2 and 3 and one change from Step 4 (power 90%). Say Steps 1 and 5 in one sentence each. Use the full 8-minute version only if the class is ahead of time.

Step 1: Frame it (1 min). *"Every sample-size calculation needs the same ingredients: what you expect in each group, how noisy the measurement is, the false-alarm risk you accept (alpha) and how sure you want to be of detecting a real effect (power). The software does the arithmetic. The ingredients are your clinical job."*

Step 2: Reproduce Paper 1 (two means, 2 min). Mean difference / two independent means: group 1 = 70 ml/kg, group 2 = 40 ml/kg, SD 35 in both, 95% confidence (alpha 5% two-sided), power 80%,

ratio 1.

Expected: **22 per group (44 total)**. Adding 10% for deaths gives 25 per group, **50 total**, exactly as in Paper 1's methods. Spreadsheet `3_Means`: B20 = 22, B21 = 25, B22 = 50.

Step 3: Our real question, mortality (two proportions, 2 min). 40% (usual care) vs 30% (restrictive), alpha 5%, power 80%, ratio 1.

Expected per group: Fleiss **356**, with continuity correction **376** (the tool may show ±1–2; Kelsey about 358). Spreadsheet `2_Proportions`: B20 = 356, B21 = 376, after 10% loss B22 = 396 per group, B23 = 792 total.

"Same clinical question, different outcome: fluid balance needed 50 patients, mortality needs almost 800."

Step 4: Change one input at a time (2 min). Ask the class to predict up or down in the chat first.

Change (two proportions, Fleiss, no correction)	n per group	Message
Baseline 40% vs 30%, power 80%	356	
Power 90%	477	more certainty costs patients
40% vs 35%	1,471	halving the difference roughly quadruples n
40% vs 25%	152	big effects need few patients, but are they realistic?
Alpha 1%	530	a stricter alarm needs more patients

Step 5: The sentence (1 min). Show the green sentence at the bottom of sheet `2_Proportions`: *"Assuming the outcome occurs in 40% of patients in group 1 and 30% in group 2, 356 patients per group give 80% power ... we will recruit 396 patients per group (792 in total)."*

Link to FLUID-ICU: its SAP (section 11) states that about 600 patients give about 80% power for an OR of 1.7 and about 55% for an OR of 1.5: *"adequately powered for a moderate association only."* That is why its adjusted CI is wide.

Part B: Comparing two groups in FLUID-ICU (03_compare.R, 6 minutes)

Before each block, ask the class for the test in the chat, using the decision table.

Block	Outcome (type)	Test	Expected output (key_results.md)
1	Age (symmetric)	Welch t-test	mean 64.6 vs 61.4 years
2–3	SOFA, lactate (skewed / score)	Mann–Whitney	SOFA 6 vs 8; lactate 2.2 vs 2.9 mmol/L (91 without lactate drop out)
4	28-day death (binary)	2×2 table + chi-square	**134/373 (35.9%) vs 45/245 (18.4%)**; RD **17.6** points (95% CI 10.7 to 24.4); RR **1.96**; p < 0.001
5	AKI (binary)	chi-square	**230/373 (61.7%) vs 71/245 (29.0%)**

Block	Outcome (type)	Test	Expected output (key_results.md)
6	Ventilator-free days (0–28, two piles)	Mann–Whitney	median **20 (0 to 28) vs 28 (22 to 28)**; p < 0.001
7	ICU stay in survivors (skewed): the common mistake	t-test vs Mann–Whitney	t-test difference in means **2.7** days vs Hodges–Lehmann **2.5** days (95% CI 1.8 to 3.2); medians **7.5 vs 4.8**

Teaching points: the test follows the outcome type, the effect is reported with a 95% CI as well as p, and every comparison here is **unadjusted** (the positive group was sicker: Table 1).

Fallback: the slide *FLUID-ICU: four outcomes, four comparisons* (all results in one table).

Part C: Models and missing data (04_models.R and 05_missing.R, with the 6.4 slides)

Run each block as you reach the matching slide. If time is short, show only the figures (they are the slides).

Slide	Script block	Expected output (key_results.md)	Say
The SAP came first	open `SAP_FLUID-ICU.md`	confounders: age, SOFA, lactate, vasopressor, ventilation, CKD; one interaction (heart failure); CC + MI	"Decided before anyone saw a result."
Crude vs adjusted	04: blocks 1–2	crude OR **2.49 (1.70 to 3.70)**, n = 618; crude in the same 527: **2.22 (1.48 to 3.39)**; adjusted **1.62 (1.02 to 2.59)**, p = 0.044, n = 527	"Confounding by indication, made visible."
Forest plot	04: blocks 3–4	the adjusted model, exposure row 1.62	"Only the exposure row is the question."
Kaplan–Meier	04: block 5	day-28 survival **81.6% vs 64.1%**; log-rank p < 0.001 (chi-square 21.4)	"Steps, ticks, gap, numbers at risk."
Cox	04: block 6	crude HR **2.17 (1.55 to 3.05)**, adjusted **1.46 (1.00 to 2.13)**, p = 0.053; PH global test p = 0.10	"0.044 vs 0.053: the same story."
(extra) Interaction	04: block 7	heart failure OR **4.39 (1.65 to 12.38)**, n = 90; no HF **1.29 (0.77 to 2.19)**, n = 437; p interaction **0.031**	"Pre-specified, plausible, needs confirmation."
Who is missing?	05: blocks 1–2	lactate missing **93/620 (15.0%)**; SOFA 5 vs 8; mortality **19.4% vs 30.6%**	"Not MCAR; MAR plausible."

Slide	Script block	Expected output (key_results.md)	Say
CC vs MI	05: blocks 3–5	Q4 adjusted OR per mmol/L: CC **1.39 (1.24 to 1.57)**, n = 527 vs MI **1.45 (1.29 to 1.64)**, n = 620; Q1 MI adjusted OR **1.70 (1.09 to 2.64)**	"Robust: the answer barely moves."

Point out in `05_missing.R` block 4: the imputation model includes the **outcome** and the predictors of missingness (SOFA, MAP, vasopressor, site), 20 imputed data sets, pooled with Rubin's rules. The density plot shows imputed lactates are lower (median 2.4 vs 2.6 mmol/L in complete cases), as expected because the missing patients were less sick.

Fallback plan

Problem	Do this
No internet (Part A)	Run Steps 2–4 in the spreadsheet (yellow cells = inputs).
R fails (Parts B–C)	Every result is on the slides, so talk through them.
`mice` slow	Skip the live run and show `cc_vs_mi.png`.
Running late	Part A: Steps 2–3 only. Part B: blocks 4 and 7. Part C: figures only.

Anticipated questions

"Which sample-size result is right, 356 or 376?" Both are standard. They use slightly different approximations. Name the method in the protocol and stay with it.

"Why did the OR go down after adjustment?" Sicker patients received more fluid and died more often. Adjustment compares patients of similar severity, so the part of the association due to severity disappears.

"The Cox p is 0.053, so there is no effect?" No. The HR (1.46) and OR (1.62) are similar with overlapping CIs, and the evidence is the same. Read the estimate and CI, not the side of 0.05.

"Isn't multiple imputation making up data?" It fills each gap many times with plausible values predicted from recorded data and widens the CI for the uncertainty. It is valid when data are MAR. Always compare with complete case.

"Why not adjust for AKI?" It happens after the fluid and may be on the causal pathway (a mediator). Adjusting for it would remove part of the effect we want to estimate. The SAP excludes it for that reason.